



ZIMBABWE

**MINISTRY OF HIGHER AND TERTIARY EDUCATION,
INNOVATION, SCIENCE AND TECHNOLOGY
DEVELOPMENT**

**HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)**

NATIONAL CERTIFICATE

IN

**MOTOR VEHICLE MECHNAICS, AUTO-ELECTRICS AND ELECTRONICS,
PRECISION MACHINING, DIESEL PLANT FITTING, MOTOR VEHICLE BODY
REPAIR AND MOTOR CYCLE MECHANICS**

SUBJECT: Automotive Engineering Drawing PAPER NO: 346/18/S04

DURATION: 3 Hours

MARCH/APRIL 2023 EXAMINATION

REQUIREMENTS

Drawing instruments, drawing papers, calculator.

INSTRUCTIONS TO CANDIDATE

Section A:	Answer ALL questions.	10 marks
Section B:	Compulsory question.	15 marks
Section C:	Answer ANY FOUR (4) questions.	40 marks
Section D:	Compulsory question.	35 marks

This paper consists of 6 printed pages.

SECTION A: Answer ALL questions.

10 marks

QUESTION 1

- 1.1 State the dimensions of an A2 size drawing sheet. (1 mark)
- 1.2 A perpendicular to the tangent at a point of tangency is called _____. (1 mark)
- 1.3 write down the following abbreviations in full;
- (i) S'FACE (1 mark)
- (ii) CH HD (1 mark)
- 1.4 Isometric and oblique drawings are _____ dimensional drawings. (1 mark)
- 1.5 Define a reducing scale. (1 mark)
- 1.6 Give any two features not normally sectioned even though it may lie in the section plane. (2 marks)
- 1.7 A protractor is used _____. (1 mark)
- 1.8 Define hatching lines. (1 mark)

420 x 595
292 x 420
A4 210 x 297

29 297
2
210
2 594
420

SECTION B:

AUTOCAD

15 MARKS

Answer the question.

QUESTION 2

(a) State the functions of the following terms with reference to AutoCAD;

(i) Snap (2 marks)

(ii) Pan (2 marks)

(iii) Zoom (2 marks)

(iv) Polar (2 marks)

(b) Write down the steps in performing the following functions as given in AutoCAD;

(i) Fillet (3 marks)

(ii) Extrude (4 marks)

SECTION C

50 marks

Answer ANY FIVE questions

QUESTION 3

Draw an internal and external arc of radii 50mm and 100mm touching on two circles of diameter 60 mm and 80mm which are 90mm apart. (10 marks)

QUESTION 4

- (a) Draw a circle of $\varnothing 70\text{mm}$ and label the chord, segment, quadrant, normal and tangent. (5 marks)
- (b) Draw a triangle ABC and label the apex, vertical angle, hypotenuse and altitude. (5 marks)

QUESTION 5

Using concentric circle method, construct the ellipse. Given that the minor axis is 70mm and major axis is 110mm. Show all construction lines. (10 marks)

QUESTION 6

Draw a circle of $\varnothing 82\text{mm}$ and construct an octagon within the circle. (10 marks)

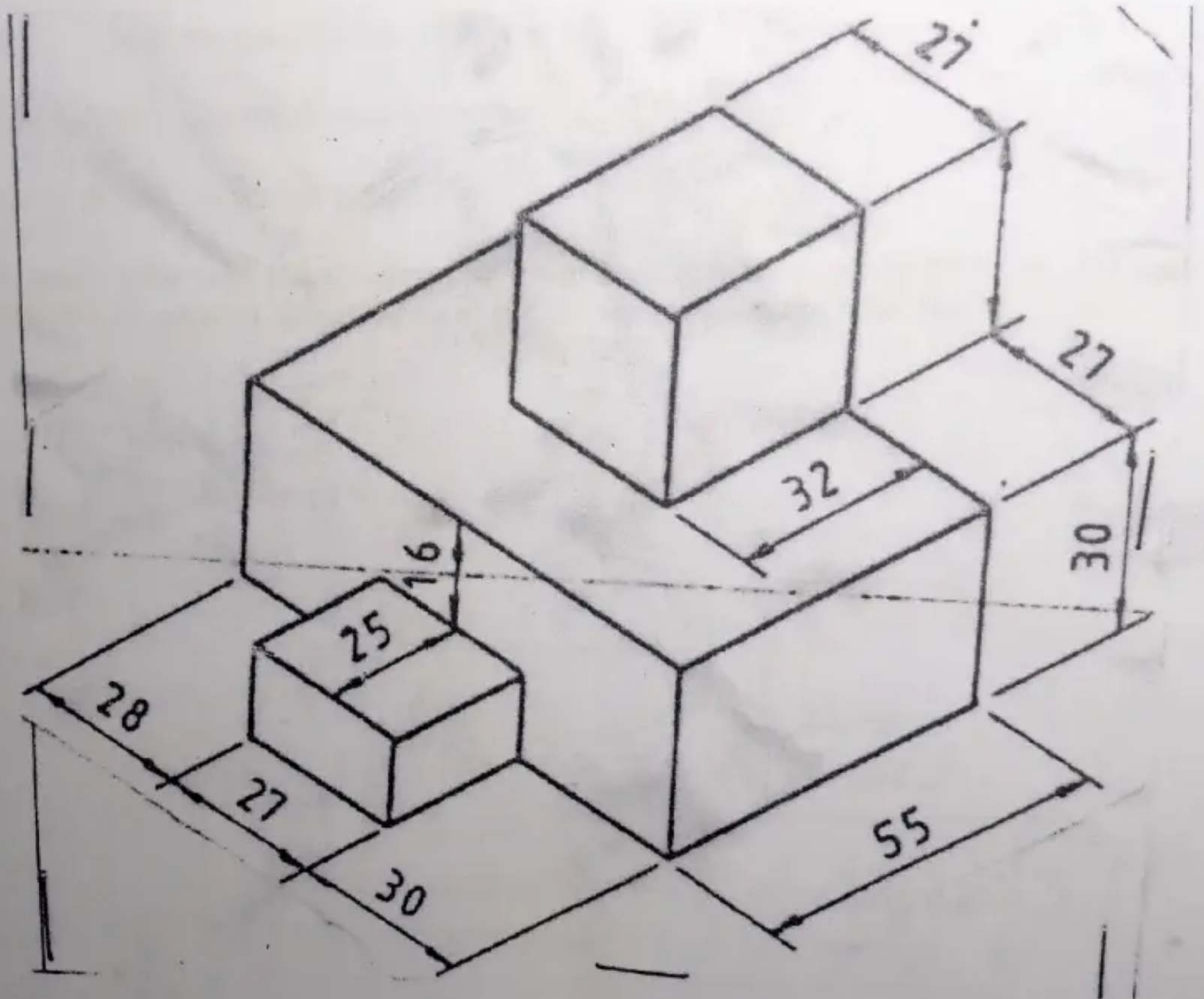
QUESTION 7

Illustrate by means of diagrams the following conventional representation;

- (a) square on end of shaft (2 marks)
- (b) diamond knurling (2 marks)
- (c) compression spring (2 marks)
- (d) break in a rod (2 marks)
- (e) break in a pipe (2 marks)

QUESTION 8

The diagram below shows a block in isometric projection. Reproduce the diagram showing all constructions. (10 marks)



Draw the following:

- In each view take the cutting plane on the centre line of the bearing. Include any six important dimensions together with a suitable title block.

